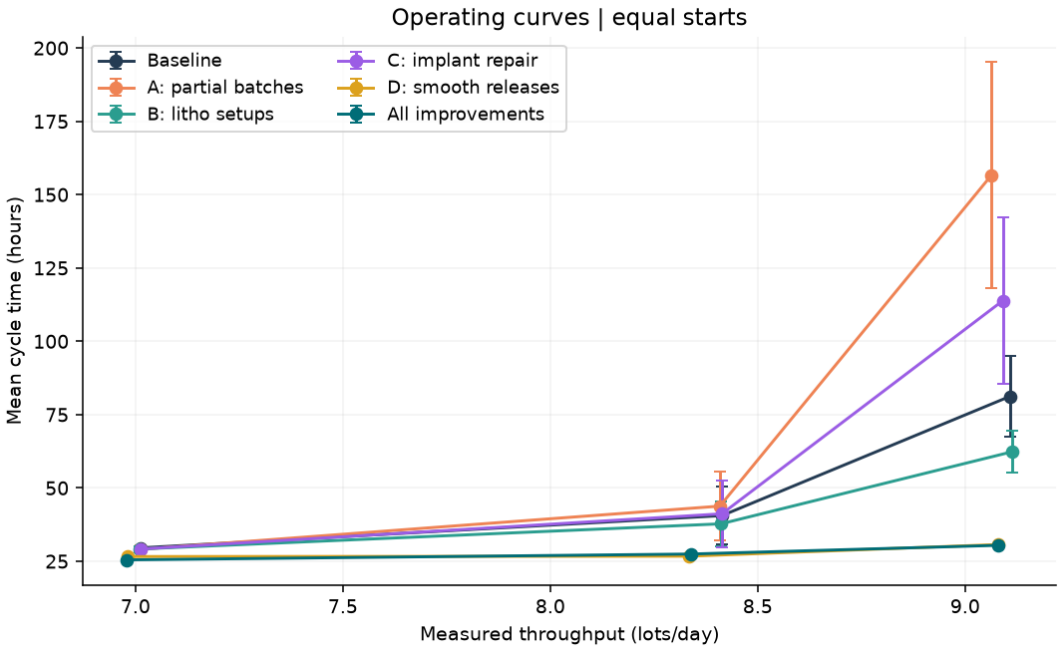
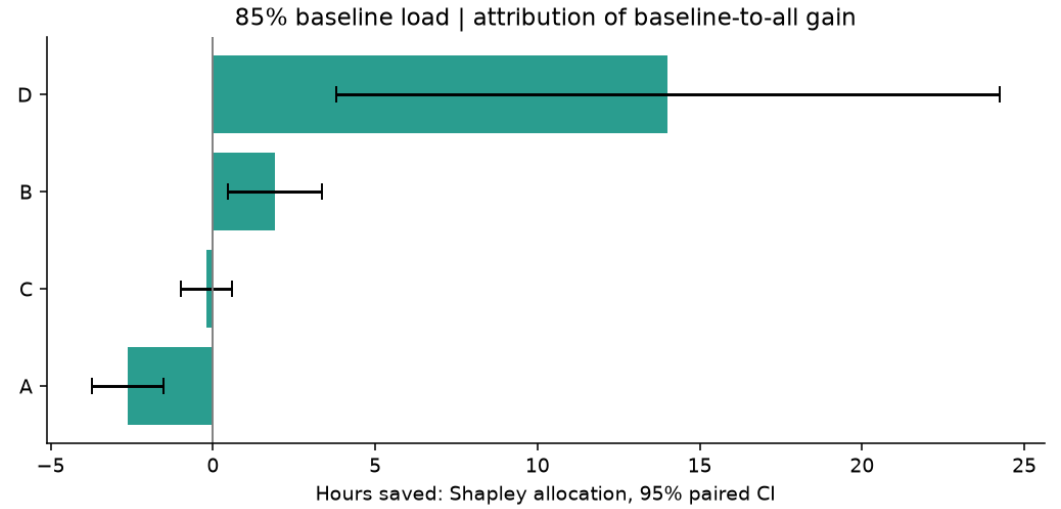
Cycle time: smooth the starts first

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At 85% baseline load and equal starts, smoothing releases reduces mean cycle time from **40.6 to 26.7 hours**. The best tested package is **BCD: 25.2 hours**. None meets the illustrative 24-hour target.

Model correction: published setups are on lithography (B); random repairs are on implant (C). The brief assigns these to the opposite stations.





Largest interaction: BD; difference-in-differences +3.29 h. Signed contributions, not physical delay shares.

The Pareto allocates the 13.1-hour baseline-to-all gain using all 16 cells. Negative values denote harmful contributions. Largest two-factor interaction: BD, difference-in-differences +3.29 h. Partial batching can lose more furnace capacity than it saves in waiting.

Decision budget and validation limits

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Decision and cost model

The 24-hour target is infeasible. D alone costs 2 illustrative units and reaches 26.72 h; BCD reaches 25.23 h. Costs A/B/C/D = 1/3/5/2 are assumptions, not prices. The dashboard recalculates the least-cost feasible package for user-supplied costs and targets, including an upper-95%-CI screen.

Experimental design

960 factorial runs cover 16 combinations, three loads and two views (equal starts and calibrated equal utilization). Measurement horizons: 365 days with five replications at 70%/85%; 730 days with 20 replications at 95%. All discard 90 warm-up days. Independent pilots, common random streams, blocked ANOVA, paired contrasts, raw/log residual checks and Shapley attribution are included.

Verification and precision

96/96 cell means meet the 5% precision criterion for the measured horizon (maximum relative half-width 4.87%). The balanced factorial uses 960 runs. Two fixed confirmation stages add 6,989 and 256 runs. Nine model tests passed. The 95% equal-starts ACD case showed rising WIP in a ten-year check, so its interval describes the finite measurement window. The 85% baseline confirmation gives 39.10 hours.

Theory and diminishing returns

VUT uses measured arrival and effective-service variability. Implant requires a two-server correction; diffusion additionally needs batch formation delay. Re-entrant dependence, operator contention and idle-time vacations explain limitations. Timeout/setup/repair sweeps are supplied. Faster repairs and shorter batch waits are not uniformly beneficial under the implemented resource schedules; do not extrapolate a monotonic benefit.

External benchmark failed

At 120-minute exponential starts, the simplified reference mode measured 13,554 minutes CT and 114.23 lots WIP versus 1,813 minutes and 15.66 lots published. The 30-run comparison uses the reported mix probabilities, simpler batching, no personnel constraints and a 22-week warm-up. BENCHMARK.md explains the capacity discrepancy and a second independent comparison.

Scope and sources

The core uses one product; a mixed-product/test-lot extension is included without test recirculation. Exact break timing, PM arbitration and interruption rules are declared modeling conventions. Optional disruption recovery, wet-clean constraints and CONWIP are outside this study. Sources: El-Khouly (2015), Chapter 4; Wang (2015), Table 4.1; McGinnis (2016); Factory Physics and Simulation (2004). Full links and parameter provenance appear in MODEL.md.